

Quantum Readout based on Silicon Photonic Integrated Circuits for Enhanced Authentication Security

G. Sarantoglou⁽¹⁾, N. Tzekas⁽¹⁾, G. Moustakas⁽²⁾, G.A. Karydis⁽²⁾, Veronika Kaminski⁽³⁾, Ekaterina Protsenko⁽³⁾, K. Gradkowski⁽⁴⁾, A. Bazin⁽⁵⁾, C. Vigliar⁽⁴⁾, A. Bogris⁽²⁾, C. Mesaritis⁽¹⁾

⁽¹⁾ Department of Biomedical Engineering, University of West Attica, Agiou Spyridonos 11243, Egaleo, Athens, Greece cmesar@uniwa.gr

⁽²⁾ Department of Informatics & Computer Engineering, University of West Attica, Agiou Spyridonos 11243, Egaleo, Athens, Greece

⁽³⁾ Department of Electrical and Photonics Engineering, Technical University of Denmark (DTU), DK 2800 Kgs. Lyngby, Denmark

⁽⁴⁾ Tyndall National Institute, Lee Maltings Complex, Dyke Parade, T12 R5CP, Corke, Ireland

⁽⁵⁾ LIGENTEC France, 224 Boulevard Kennedy, 91100 Corbeil-Essonnes, France

Abstract In this work, we present a quantum-driven physical unclonable function based on a silicon nitride photonic integrated chip, offering enhanced authentication security against eavesdropping. The maximally mixed state property conceals the hardware fingerprint, while a cloning probability as low as 10^{-14} provides secure authentication. ©2026 The Author(s)

Introduction

In the modern landscape of rapidly increasing data exchange, privacy and security are critical concerns. Authentication typically relies on digital keys stored in non-volatile memory, which are vulnerable to hardware attacks and often require power-intensive protection [1]. Physical unclonable functions (PUFs) address this by exploiting inherent fabrication variations (FVs) to generate unique challenge–response pairs (CRPs) [2]. Common PUFs include SRAM, arbiter circuits, ReRAM devices [2], optical systems [3], and more recently photonic integrated circuits (PICs) [4], [5], [6]. A major challenge is vulnerability to machine learning attacks, where collected CRPs enable accurate device modelling [2]. While increasing FV complexity is one approach, an alternative is to prevent CRP acquisition altogether by using quantum-based protection.

A practical quantum realization consists of a scheme, where I/O signals are quantum states and the PUF is a classical process, known as quantum readout (QR) [7]. This approach is impractical for fermionic platforms due to low temperature requirements, but highly compatible with photonics that enable room-temperature operation and simpler hardware [8]. QR has so far been demonstrated only in bulk optics [9], with QR-PICs remaining unexplored [10]. In this work, we present an experimentally supported theoretical investigation of QR-PICs based on a 6×6 rectangular mesh for secure authentication [11]. FVs arise mainly from waveguide side-wall roughness, that leads to random refractive index permutations [6]. It is shown, that merging this system with single-photon maximally mixed states (MMS) conceals FVs. First, the system is

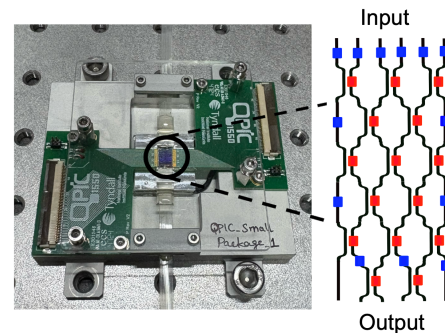


Fig. 1: The experimental PIC (left) and the schematic of the circuit (right).

validated as a PUF using classical inputs. Next, extracted parameters are used to simulate the quantum protocol. The QR-PIC presents false acceptance rate (FAR) of 10^{-14} and false rejection rate (FRR) of 10^{-6} demonstrating strong authentication security.

Classical Photonic Integrated PUF

The experimental setup consists of a 6×6 SiN interferometric mesh (Fig.1) in a rectangular topology [11], controlled by 28 thermo-optic phase shifters (TOPS). The mesh performs a unitary mapping configured by the TOPS [11]. A 5 dBm CW C-band laser is injected into the first port, and the response is considered the power distribution across the PIC's ports. This profile, determined by the mesh transfer function, is transformed via a digital process - random projection, quantization and binarization - to a 256-bit response [6]. The applied TOPS voltages affect the response by changing the power distribution. Device characterization shows propagation losses $\eta_{dB} = 11$ and nullification voltages that follow an approximately uniform distribution (Fig. 2).

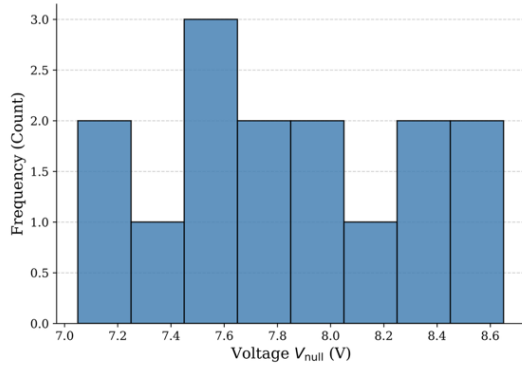


Fig. 2: The uniform distribution the 15 nullification points V_{null} of the MZIs

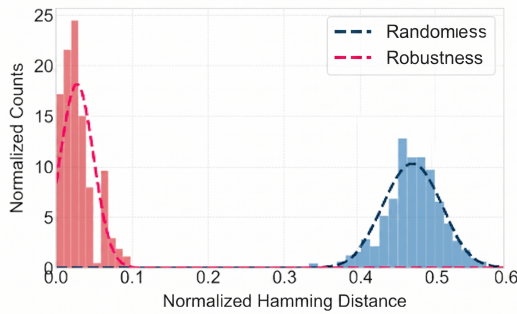


Fig. 3: Randomness and Robustness PDFs for the classical PIC based PUF.

Phase shifter values are given by $\phi(V) = \pi(V^2 - V_{null}^2 + V_{\pi}^2)/V_{\pi}^2$, $V_{null} \sim U(7, 8.6)$, $V_{\pi} = \mathbb{E}[V_{null}]$. The offsets introduced by V_{null} points constitute the hardware fingerprint due to FVs. For validation, 30 challenges were applied, each defined by a TOP voltage vector $V \in \mathcal{R}^{10}$, where voltages are set within $U(5V, 10V)$ around the null points. Each challenge produces a 256-bit response. Randomness is evaluated using the normalized Hamming distance (NHD) between responses, yielding a Gaussian distribution with mean 46.25% and standard deviation 2.5%, indicating high entropy (Fig.3).

In practice, the response deviates from the ideal due to effects like thermal crosstalk, finite DAC resolution, thermal drifts and polarization drifts [12]. A Peltier controller stabilizes the temperature at $22 \pm 0.1^\circ\text{C}$ to account for global thermal drifts. The maximum power drift among 66 realizations, for 30-minute intervals (well above the thermal time constant) and a fixed transformation, was 0.2 dB showcasing insignificant variations. The resulting NHD distribution for robustness has mean 2.6% and standard deviation 2.03%, demonstrating high stability (Fig.3).

Photonic Integrated Chip as Quantum Readout

The protocol for QR-PIC is shown in Fig. 4. We consider a passive eavesdropper (Eve) who monitors I/O ports, while active man-in-the-middle

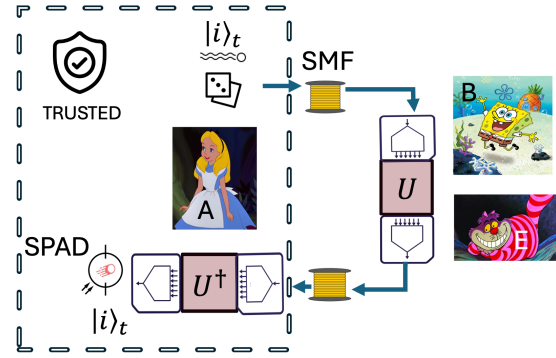


Fig. 4: The QR-PIC protocol

attacks are treated in the Discussion section. The protocol is based on Alice and Bob communication with multiple single-photon trials. It is pre-supposed that Alice knows Bob's passive offsets via an enrolment stage.

In each trial, Alice generates a random index $i \in \{0, \dots, D\}$ uniformly using a secure random number generator [13] and prepares a Fock state $|1\rangle$ via heralding [14]. For Alice, the state is encoded into a temporal qudit $|i\rangle$ and transmitted to Bob via a single-mode fiber [15]. Bob converts the temporal qudit to a spatial mode using a photonic transcoder - delay spirals and optical switches [16]. A FV-dependent unitary transformation $U_B = U_{D \times D}$ (D is the number of ports) is applied via the PIC, thus encoding the hardware fingerprint to the pure output state $U_B|i\rangle$. The response is encoded as a temporal qudit [16] and transmitted back to Alice. Upon reception, Alice, applies the inverse transformation $U_A = U_{D \times D}^\dagger$ by programming her own PIC to match Bob's configuration and recovers $U_A U_B|i\rangle = |i\rangle$. The output is detected at the SPAD in the i -th time bin via a tagger. Deviations from $|i\rangle$ appear as mis-clicks and indicate either an incorrect transformation or detector noise. The probability of a correct outcome given a click is obtained via Bayes rule, $P(C|K) = P(C)P(K|C)/P(K)$ (1). Given $F = |U_A U_B|^2$, $\text{tr}(\cdot)$ its trace, η the transmission efficiency and d the dark count probability, the probability of a correct click is $P(C) = \eta \text{tr}(F) D^{-1} + d$, weighted by the absence of dark count clicks at wrong bins $(1-d)^{D-1}$, while the total click probability is $P(K) = \eta + Dd$. Also $P(K|C) = 1$. The error probability is $P_e = 1 - P(C|K)$.

$$P(C|K) = \frac{(\eta \text{tr}(F)/D + d)(1-d)^{D-1}}{\eta + Dd} \quad (1)$$

Alice knows that the pure state $|i\rangle$ is sent without performing a measurement and therefore describes the state as pure throughout propagation in the PIC. Eve (and Bob), on the other hand,

must perform a measurement to infer the state. This asymmetry in available information implies that, from Eve's perspective, the input to the PIC prior to any measurement is described by an MMS, $\rho_{in} = I/D$, due to the uniform distribution over i [17]. Here, ρ denotes the density matrix and $I_{D \times D}$ the identity matrix. Since unitary transformations preserve an MMS, the output ensemble remains $\rho_{out} = I/D$ for Eve, independent of the applied transformation, thereby concealing U [17].

A full session works as follows: Alice first defines an error-correction number N_T and a desired total number of clicks, which marks when the session ends. Then multiple trials are performed until the stop condition is met. Afterwards, Alice holds a mapping between the expected and the received clicks. Up to N_T errors are acceptable to account for non-idealities, otherwise the applicant is rejected. The probability for less than N_T errors given N_c clicks is:

$$P(n \leq N_T) = \sum_{n=0}^{N_T} \binom{N_c}{n} p_e^n (1 - p_e)^{N_c - n} \quad (2)$$

From this equation, error rates can be calculated as: $FRR = 1 - P(n \leq N_T)$ and $FAR = P(n \leq N_T)$.

Numerical Results

Numerical evaluation uses the parameters of section II and the model in [18] for the 6×6 unitary. Loss budget consists of 11 dB per PIC (22 dB), 25% SPAD efficiency [19] (6 dB), 4 dB fiber loss (20 km total, 0.2 dB/km), and dark counts of 10^{-6} (1000 counts/sec, 1 ns gate), yielding a total loss of 32 dB. Phase offsets at transcoders are assumed known and compensated by Alice, while Bob's unitary is defined by FV-induced V_{null} variations at $V = V_{\pi}$.

Non-idealities (i.e. thermal crosstalk, DAC resolution, polarization/thermal drift) lead to $U_A \neq U^\dagger, U_B \neq U$. Experimental drifts are transformed to probability noise via normalization of the power profile of the 66 recordings, leading to $\sigma = 2.4 \times 10^{-3}$. This is added to F as Gaussian noise $\mathcal{N}(0, \sqrt{2}\sigma)$, where $\sqrt{2}$ accounts for the two PICs. Monte Carlo analysis (5000 runs) shows that FRR decreases with increasing threshold (Fig. 5). $FRR \approx 3 \times 10^{-6}$ for $N_T = 10$ and 115 clicks.

The effect of FVs is evaluated through the FAR. 5000 Monte Carlo realizations of U_E are computed, for PICs possessed by Eve and originating from the same wafer as Bob's. Due to FVs, the realized transformations deviate from the enrolled U_A resulting in imperfect inversion $F = U_A U_E \neq I$. In Fig. 6, increasing N_T raises FAR due to relaxed acceptance criteria, whereas increasing clicks results in FAR reduction due to accumulating errors. For $N_T = 10$, and 115 clicks, FAR reaches close to the excellent value of 10^{-14} .

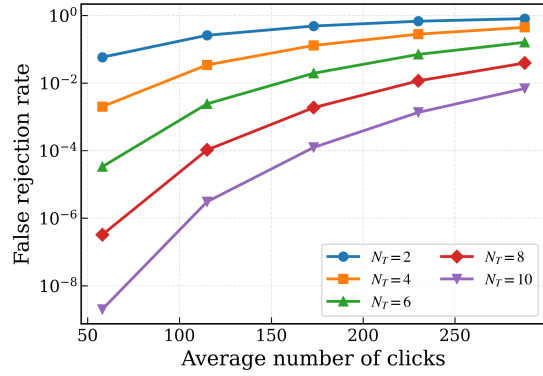


Fig. 5: FRR versus the number of shots and fixed errors

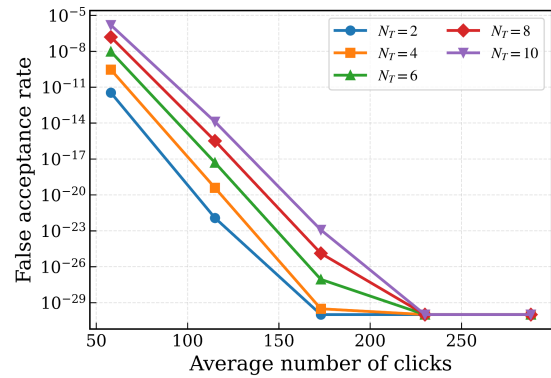


Fig. 6: FAR versus the number of shots and fixed errors

Discussion

A passive eavesdropper gains no information due to the MMS property. A stronger threat is a man-in-the-middle attack, where Eve attempts to reconstruct the unitary ($D^2 - 1$ parameters, e.g., 35 for $D = 6$) [19] by actively sending photons to Bob's chip during a session. In a typical session (~100 clicks), this yields less than 3 samples per parameter, leading to inadequate estimation accuracy. After each session, Alice and Bob modify the TOP settings, changing the unitary. Therefore, reliable reconstruction is infeasible.

Conclusion

We present a quantum readout protocol for secure authentication supported by simulations based on experimentally derived parameters from a 6×6 SiN rectangular mesh. The photonic integrated chip functions as both a classical PUF and quantum readout primitive. Experimental investigation for the classical mode of operation, reveals 45% entropy and 2.6% robustness, validating its complexity and stability. In the quantum regime, single-photon inputs and SPAD detection exploit the MMS property to conceal the unitary transformation from passive eavesdroppers. Device cloning is evaluated with experimentally derived voltage biases originating from FVs. Under realistic impairments, including dark counts, the system achieves $FRR \approx 10^{-6}$ and $FAR \approx 10^{-14}$, demonstrating outstanding security.

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